

A Project Report
Books Inventory System



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Project Overview

Project Name- Book Inventory System

Language- Perl

Hardware & Software Requirement

Software- Perl Environment

Hardware- Windows, Mac, Linux OS

Project Description

The goal of this project is to maintain books inventory and order new books if stock of the books gets less, this system can be very helpful in various places such as a book shop, a library, reading centers of various institutes.

The goal is to help the user keep a database of the books in stored format, to maintain efficiency and enhance the workflow of a book environment(book store, library etc).

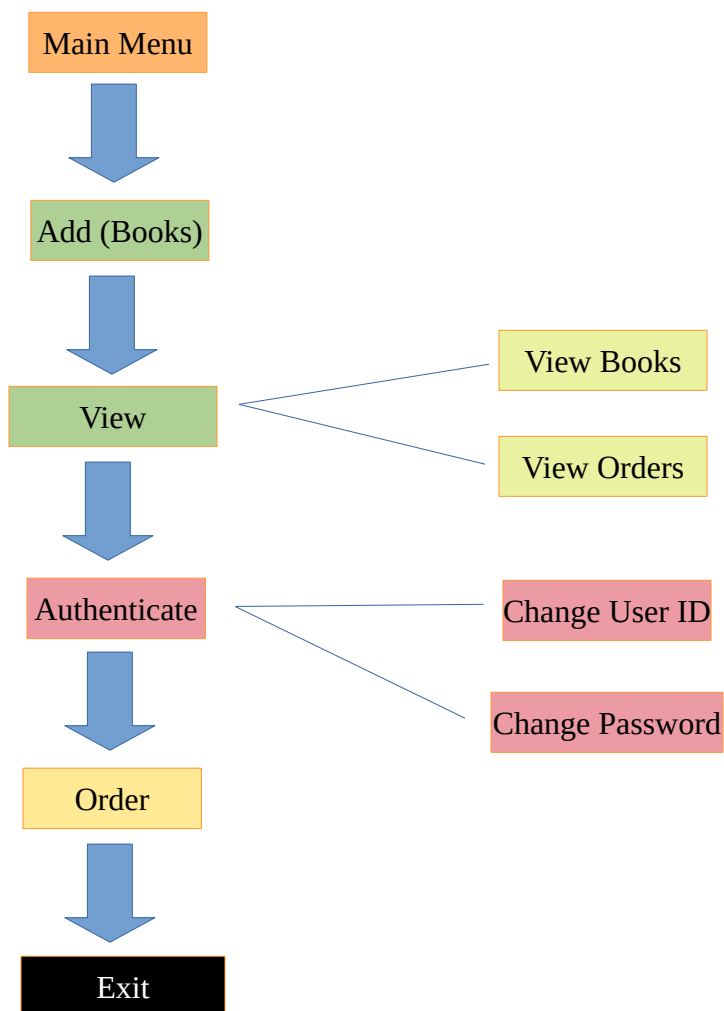
The back-end relies on flat-file storage (such as text or CSV files) in this project I have used .txt file to save and retrieve records, ensuring the system remains lightweight and easily portable. Perl's strong text processing capabilities make it well-suited for this task.

Flow Chart

The Book store management process flowchart aims to give a clear understanding to the admin about the flow of process, the project goes through and what are the functions which the admin can do to manage the book store.

Here, the main menu contains the following attributes which are – Add Books, Search Books, Delete, Order Books, View Records.

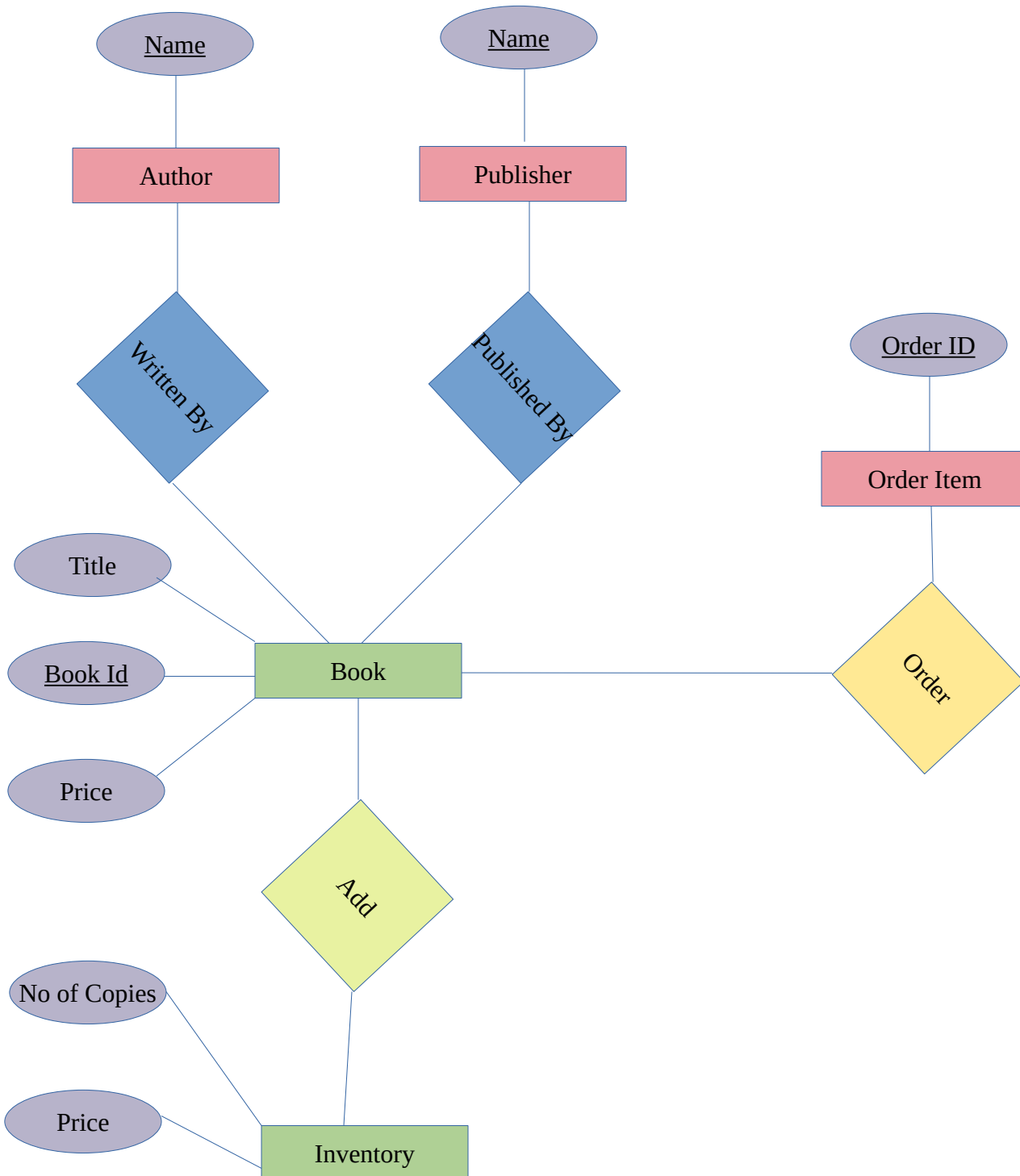
Inventory functions which admin can perform are- adding new books, search books & order, delete order & record, View Record.



E-R Diagram

Entity Relationship Diagram shows the relationship between the attributes and objects.

Here, the primary keys are – Author Name, Book Id, Publisher Name & Order Id.



Future Growth of the Project

There are few more functions which can be added in the future to the project.

1. Update and Delete Book Records

- Updating the existing book details (e.g., price, quantity, title) through the system can also be added in the future. Also, deletion of books functionality will further complete the whole data base management system.

2. Search and Filter

Currently, the user can view books and order records and implementing keyword-based search (by title, author, subject) will further upgrade the functionality. Searching of books can be done by adding filters for category such as availability, price range, etc.

3. User Management

Currently the project has only one user login implemented and allowing multiple user roles like Admin, Staff, Customer will further improve the accessibility of the system by multiple users.

4. Inventory Alerts

Inventory alerts are automated notifications or visual indicators that help staff or admins stay informed about the stock levels of books. The goal is to avoid stock outs (when a book is out of stock) or overstocking (when too many unsold books accumulate), ensuring smooth inventory management and timely restocking.

5. File Management Integration

Currently, the project database connectivity is with the notepad. Adding the connectivity with Excel File(.csv) can enable the data analytics functions to the system.

These future improvements can enhance the usability, scalability, performance, and overall value of the book management system.

Introduction to Perl Language

Perl is a high level interpreted and dynamic programming language.

- Perl supports both the Procedural and Object-Oriented programming.
- Perl is a lot similar to C syntactically and is easy for the users who have knowledge of **C**, **C++**.
- Perl code is typically interpreted, meaning that it is executed directly by the computer without the need for compilation. This can make development faster and more flexible, but can also lead to slower execution times for large programs.

Advantages

- It is very efficient in text-manipulation i.e. Regular Expression. It also provides the socket capability.
- It is free and a Open Source software.
- Flexibility: Perl is a flexible language that allows programmers to solve problems using a variety of approaches, making it easy to write code quickly and efficiently.
- Regular expressions: Perl has excellent support for regular expressions, which makes it ideal for text processing and data manipulation tasks.
- Large library of modules: Perl has a vast library of modules that can be used to extend its functionality, making it easy to add complex features to your programs without having to reinvent the wheel.
- Cross-platform support: Perl runs on a wide range of platforms, including Windows, Linux, macOS, and more.

Disadvantages

- Programs runs slowly and program needs to be interpreted each time when any changes are made.
- In Perl, the same result can be achieved in several different ways which make the code untidy as well as unreadable.
- Usability factor is lower when compared to other languages.
- Complex syntax: Perl's flexibility can also be a disadvantage, as it can lead to code that is difficult to read and maintain, especially for developers who are new to the language.
- Performance: Perl is an interpreted language, which means that it can be slower than compiled languages like C or Java for large or computationally intensive programs.
- Legacy code: Because Perl has been around for over 30 years, there is a lot of legacy code written in earlier versions of the language that may not be compatible with newer versions.
- Security: Perl has some security concerns, such as the possibility of code injection attacks if the input data is not properly sanitized.

Applications

- Web development: Perl is commonly used for server-side web development, with popular frameworks such as Catalyst and Dancer.
- Text processing: Perl's powerful regular expression support makes it ideal for text processing tasks.
- System administration: Perl is often used for system administration tasks, such as automating system backups, managing configuration files, and monitoring system resources.

- Network programming: Perl can be used for network programming, with modules available for tasks such as sending email, communicating with web services, and building network protocols.
- Bio-informatics: Perl is widely used in bio-informatics, with popular modules such as Bio-Perl providing a powerful toolkit for biological data analysis and manipulation.
- GUI development: Perl can be used for GUI development with toolkits such as Perl/Tk, allowing developers to create cross-platform desktop applications.

Project Code

```
$filename = "Books_Inventory.txt";
@records;
open $fh, '<', $filename;
$header=<$fh>;
chomp $header;
@headers=split(',', $header);
while ($line = <$fh>)
{
    chomp $line;
    @fields=split(',', $line);
    %record;
    for $i(0..$#headers)
    {
        $record{$headers[$i]}=$fields[$i];
    }
    push @records, \%record;
}
```

```

close $fh;

$orderfilename = "Books_Orders.txt";
@records;
open $fh, '<', $filename;
$header=<$fh>;
chomp $header;
@headers=split(',', $header);
while ($line = <$fh>)
{
    chomp $line;
    @fields=split(',', $line);
    %record;
    for $i(0..$#headers)
    {
        $record{$headers[$i]}=$fields[$i];
    }
    push @records, \%record;
}
close $fh;

```

```

sub Bookshoplogin
{
    open($fh, '<', 'Login_Details.txt');
    chomp($saved_id = <$fh>);
    chomp($saved_password = <$fh>);
    close($fh);
    print ("-----Login----- \n");
}

```

```

print ("Enter the Login ID: \n");
chomp($login_id=<STDIN>);
print ("Enter the Password: \n");
chomp($login_password=<STDIN>);

if($login_id eq $saved_id && $login_password eq $saved_password)
{
    print ("Login Successful \n");
    mainmenu();
    return;
}
else
{
    print ("Incorrect Login ID And Password. Try Again \n");
    Bookshoplogin();
}
}

sub mainmenu
{
    print "\n === Welcome To Book Shop === \n";
    print "Choose from the Options Below :\n";
    print "1. Add \n";
    print "2. Order \n";
    print "3. Change Authentication \n";
    print "4. View Details \n";
    print "5. Exit \n";
    print "Enter Your Choice: ";

```

```
chomp ($input = <STDIN>);

if ($input eq '1')
{
    add();
}
elseif ($input eq '2')
{
    Order();
}
elseif ($input eq '3')
{
    authenticate();
}
elseif ($input eq '4')
{
    View();
}
elseif ($input eq '5')
{
    exit();
}
else
{
    print ("Invalid choice. Please Select a Valid Option.\n");
    mainmenu();
}
```

```

}
sub add
{
    print ("\n Enter Book Title: ");
    $title = <STDIN>;
    print ("\n Enter Book ID: ");
    $bookid = <STDIN>;
    print ("\n Enter Author Name: ");
    $author = <STDIN>;
    print("\n Enter Publisher Name: ");
    $publisher = <STDIN>;
    print ("\n Enter Price: ");
    $price = <STDIN>;
    print ("\n Enter No of Copies: ");
    $quantity = <STDIN>;

    chomp($title, $bookid, $author, $publisher, $price, $quantity);
    open($fh, '>>', $filename);
    print $fh join(',', $title, $bookid, $author, $publisher, $price,
$quantity) . "\n";
    close($fh);
    print ("\n New Book Data Added successfully to $filename \n");
    mainmenu();
}
sub View
{
    while (1)
    {

```

```

print "Press 1 To View All the Books Records \n";
print "Press 2 To View Order Records \n";
print "Press 0 For Main Menu \n";
print "Enter Your Choice: \n";

chomp ($input = <STDIN>);

if ($input eq '1')
{
    open $fh, '<', $filename;
    printf "%-20s %-6s %-20s %-30s %-7s %-8s\n",
        "Title", "ID", "Author", "Publisher", "Price", "Quantity";
    print "-" x 100 . "\n";
    while($line=<$fh>)
    {
        chomp($line);
        ($title, $bookid, $author, $publisher, $price, $quantity)
= split /,/, $line;
        printf "%-20s %-6s %-20s %-30s %-7s %-8s\n",
            $title, $bookid, $author, $publisher, $price, $quantity;
    }
    close $fh;
}
elsif ($input eq '2')
{
    open $fh, '<', $orderfilename;
    printf "%-20s %-6s %-20s %-30s %-7s %-8s\n",

```

```

        "Name", "Order Id", "Book Id", "Price", "Quantity",
"Warehouse";
    print "-" x 100 . "\n";
    while($line=<$fh>)
    {
        chomp($line);
        ($name, $orderId, $book_id, $amount, $bookquantity,
$selected_warehouse) = split /,/, $line;
        printf "%-20s %-6s %-20s %-30s %-7s %-8s\n",
            $name, $orderId, $book_id, $amount, $bookquantity,
$selected_warehouse;
    }
    close $fh;
}
elseif ($input eq '0')
{
    mainmenu();
}
else
{
    print ("Invalid choice. Please Select a Valid Option.\n");
    View();
}
}
}
sub Order
{
    print ("Please Enter Your Name: \n");
    $name = <STDIN>;

```

```
print ("Entered Name is: $name \n");
print ("Please Enter The Book ID: \n");
$book_id = <STDIN>;
chomp $book_id;
print ("Entered Book ID is: $book_id \n");
open $fh, '<', $filename;
$found=0;
while($line = <$fh>)
{
    chomp $line;
    @fields = split /,/, $line;
    if ($fields[1] && $fields[1] eq $book_id)
    {
        print ("Book ID Found \n");
        print ("$line \n");
        $found = 1;
        last;
    }
}
close $fh;
if (!$found)
{
    print "Book Not Available.\n";
    mainmenu();
}
print ("Enter the Quantity: \n");
$bookquantity = <STDIN>;
print ("Entered Quantity is: $bookquantity \n");
```

```
print ("----- Warehouse ----- \n");
```

```
while(1)
```

```
{
```

```
print "Press 1 For ABC Warehouse \n";
```

```
print "Press 2 For CNBC Warehouse \n";
```

```
print ("Please Select the Warehouse: \n");
```

```
chomp ($input = <STDIN>);
```

```
if ($input eq '1')
```

```
{
```

```
    $selected_warehouse="ABC Warehouse";
```

```
    print ("Your Selected Warehouse is: ABC Warehouse \n");
```

```
    last;
```

```
}
```

```
elsif ($input eq '2')
```

```
{
```

```
    $selected_warehouse="CNBC Warehouse";
```

```
    print ("Your Selected Warehouse is: CNBC Warehouse \n");
```

```
    last;
```

```
}
```

```
else
```

```
{
```

```
    print ("Please Enter the Correct Choice \n");
```

```
}
```

```
}
```

```

$orderfilename = "Books_Orders.txt";
open(my $fh, '>>', $orderfilename);
print $fh "Selected Warehouse: $selected_warehouse\n";
close $fh;

print ("Enter Book Amount For 1 Unit \n");
$bookamount = <STDIN>;
$amount=$bookquantity*$bookamount;

print ("Total Amount $amount \n");
$orderID = int (rand(20));
print ("Thank You $name Your Order has been Placed
Successfully. \n");
print ("Please Note Down The Order Number: $orderID for
Futher Reference. \n");
orderdata();

sub orderdata
{
    $orderfilename = "Books_Orders.txt";
    chomp($name, $orderID, $book_id, $amount, $bookquantity,
$selected_warehouse);
    open($fh, '>>', $orderfilename);
    print $fh join(',', $name, $orderID, $book_id, $amount,
$bookquantity, $selected_warehouse) . "\n";
    close($fh);
    mainmenu();
}
}

```

```

sub authenticate
{
    print("Enter New User ID: \n");
    chomp($loginid=<STDIN>);
    print("Enter New Password: \n");
    chomp($loginpassword=<STDIN>);
    open($fh, '>', 'Login_Details.txt');
    print $fh "$loginid\n$loginpassword\n";
    close $fh;
    print("User ID and Password Changed");
}

Bookshoplogin();

sub exit
{
    exit(0);
}

```

Output (Screenshots)

First, the user needs to enter the Login ID And Password to access the menu.

```

PS C:\Users\aaakas> cd C:\Users\aaakas\OneDrive\Desktop
PS C:\Users\aaakas\OneDrive\Desktop> perl Bookshopmanagement.pl
-----Login-----
Enter the Login ID:

```

After entering the Login id and Password the user is granted access to the menu.

```
PS C:\Users\alakas> cd C:\Users\alakas\OneDrive\Desktop
PS C:\Users\alakas\OneDrive\Desktop> perl Bookshopmanagement.pl
-----Login-----
Enter the Login ID:
Admin
Enter the Password:
Admin
Login Successful
```

Menu

```
=== Welcome To Book Shop ===
Choose from the Options Below :
1. Add
2. Order
3. Change Authentication
4. View Details
5. Exit
Enter Your Choice: █
```

Adding New Book to the inventory

```
Enter Your Choice: 1

Enter Book Title: Linux

Enter Book ID: B0012

Enter Author Name: xyz

Enter Publisher Name: mnp

Enter Price: 250

Enter No of Copies: 7

New Book Data Added successfully to Books_Inventory.txt
```

Changing User Id And Password

```
Enter Your Choice: 3
Enter New User ID:
Admin123
Enter New Password:
Admin123
User ID and Password Changed
```

Ordering new book for the inventory

```
Enter Your Choice: 2
Please Enter Your Name:
Aakash
Entered Name is: Aakash

Please Enter The Book ID:
B0012
Entered Book ID is: B0012
Book ID Found
Linux,B0012,xyz,mnp,250,7
Enter the Quantity:
6
Entered Quantity is: 6

----- Warehouse -----
Press 1 For ABC Warehouse
Press 2 For CNBC Warehouse
Please Select the Warehouse:
1
Your Selected Warehouse is: ABC Warehouse
Enter Book Amount For 1 Unit
250
Total Amount 1500
Thank You Aakash
Your Order has been Placed Successfully.
Please Note Down The Order Number: 9 for Futher Reference.
```

Viewing Books Record currently in the Inventory

```
Enter Your Choice:
1
```

Title	ID	Author	Publisher	Price	Quantity
Mathematics	B001	R.D. Sharma	Dhanpat Rai Publications	550	30
Science	B002	Lakhmir Singh	S. Chand Publishing	480	25
Social Science	B003	Arjun Dev	NCERT	180	40
English Literature	B004	NCERT	NCERT	100	35
Hindi Vyakaran	B005	Dr. Vasudevanand	Saraswati House	220	20
Geography	B006	Majid Husain	NCERT	150	28
Physics	B007	Pradeep Publications	Pradeep Publications	620	15
Chemistry	B008	O.P. Tandon	GRB Books	750	18
History	B009	Romila Thapar	NCERT	140	22
Biology	B010	Trueman's	Trueman Book Company	780	12
Linux	B0012	xyz	mnp	250	7

Viewing Books Order Records

Name	Order Id	Book Id	Price	Quantity	Warehouse
Name	Order Id	Book Id	Price	Quantity	Warehouse
Aakash	1	B001	550	1	ABC Warehouse
Aakash	19	B001	3850	7	ABC Warehouse
Aakash	1	B004	600	6	CNBC Warehouse
Aakash	4	B006	150	1	CNBC Warehouse
Aakash	9	B0012	1500	6	ABC Warehouse

Conclusion

The Books Inventory System developed using the Perl programming language provides a functional and efficient solution for managing a small to medium-sized book inventory. The system has the inventory management functionalities, including:

- **Adding New Books:** Users can seamlessly add new books to the inventory database, ensuring the collection remains up to date.
- **Ordering Books:** The system allows users to place orders for new stock, with each order being recorded and stored in the database for tracking and future reference.
- **Viewing Records:** Comprehensive viewing capabilities are provided to review both current inventory and ordered books, enhancing transparency and control.
- **User Credential Management:** Users can change their user ID and password, adding a layer of personalization and security to the system.

This Perl-based system demonstrates the flexibility and utility of scripting languages in building reliable and straightforward inventory solutions. While it serves its purpose effectively, it also offers a strong foundation for further enhancements, such as integrating a user-friendly GUI or implementing advanced search and filtering features. Overall, it streamlines essential book inventory processes and is a valuable tool for managing book data efficiently.